Does Old Age Security(OAS) cause a decline in labour participation rates?

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In this paper we determine whether or not OAS has a causal impact on labour force participation rate. Using the Differences in Differences model we use the eligibility of the individual and their age with factors like gender, income and education to determine the effect on the labour force participation rate. In our analysis, it was found that there is insignificant evidence of the effect of OAS on participation in the labour force.

I. Background

To determine eligibility for our project, we checked the requirements for OAS. The main requirement is that you must be 65 years old or older, resided in Canada for 10 years minimum, and be either a Canadian citizen or a legal resident. A legal resident includes Temporary or Permanent residents.

Another factor to consider is the income of an individual. If an individual reaches 90,000 or more in income during retirement, the individual is subject to a claw back meaning you earn less then the individuals earning less than 90,000. Once income is at 150,000 the OAS is zero.

On top of the OAS some another portion of income for retirees is the CPP(Canadian Pension Plan), this is 33% of an individuals total life time earnings that went into CPP, an individual is able to withdraw from this fund at age 65.

II. Data

In this paper the data used is from the 2021 Canada Census. We filtered this dataset by individuals in the age groups 60-64 and 65-69 and did necessary cleaning. This is important as it gives us the data from 5 years before OAS starts and 5 years after. This will be important when used for our DD analysis. To determine whether or not an individual was eligible for OAS, there were two groups for this.

1) If an individual immstatus(Immigration Status) is Non-immigrant, meaning they are Canadian Born, they are eligible for OAS.
2) If an individual immstatus is Immigrant, and they have been in Canada for 10 or more years, they are eligible.

To determine eligibility for the Immigrant population, I took the yrim(Year of Immigration) and subtracted it with the year of the census(2021). If individual is Immigrant & has $years\_In\_Canada$ >= 10, they are eligible for OAS.

The main variables we are looking into for our analysis will be lfact(Labour Force Status), agegrp(Age), Gender, pr(Province), marsth(Marital Status), HHInc_AT (Household Income After Taxes), TotInc_AT(Total Personal Income After Taxes) & hdgree(Education Status). Gender is an important variable as Male vs Female workings may differ and one may be more likely to change results over the other. Province is used as the differences between the labour markets across Canada. Marital Status is also important as a spouses decisions may influence the others participation. For example, if one spouse earns enough, the other may retire because of income, not because of OAS. Education is important as there may be factors such as educated workers may work for a longer time due to work that is less physically demanding. Household income is important because the top earners may have earned enough to save over their life and be able to retire regardless of OAS support. To control for the clawback we also add in the Personal Income to see how it changes for the individuals earning more than 90,000 and ones less than.

TABLE 1—SUMMARY STATISTICS FOR VARIABLES OF INTEREST

Statistic	Mean	St. Dev.	Min	Max
Labour Force Status	0.44	0.50	0	1
Age 65+	0.46	0.50	0	1
OAS Eligible	0.81	0.39	0	1
Gender (Male = 1)	0.48	0.50	0	1
Married	0.61	0.49	0	1
HH Income (< 90000 = 0)	0.36	0.48	0	1
Per Income(< 90000 = 0)	0.15	0.36	0	1
University Degree	0.35	0.48	0	1

Sample Size of N = 126,446
Used 90,000 as the point for both Income

III. Methodology

To determine the casual impact of OAS on labour force participation our method will include using the Differences in Differences model. This approach will compare the change in the labour force participation for individuals who meet the eligibility for OAS against those who do not. We use the age as a treatment because at 65 is when OAS starts and by comparing the change before and after OAS we can see the impact. Having OAS eligibility as our groups allows us to split of the effect of OAS from the effect of aging. The basic regression formula we will be using for this Differences and Differences model is:

$$Y_i = \alpha + \beta_1 D_i + \beta_2 T_i + \beta_3 (D_i \times T_i) + \varepsilon_i$$

Setting up the potential outcomes model to determine our setup.

- Y_i: Labour Force Participation

  - 1 : in labour force
  - 0 : out of labour force

- D_i: Eligibility for OAS
  - 1 : Eligible: Canadian born or immigrant in Canada for 10 or more years
  - 0 : Ineligible: Immigrant in Canada less than 10 years

- T_i: Age group
  - 1: 65–69
  - 0: 60–64

- X_i: Controls (Education, Income, Gender, etc.)

The assumptions required to interpret the result as causal will requires 1 assumptions;

- Parallel Trends Assumption: We are assuming that if there is no OAS, the labour force participation rate for the ineligible group would have have same trend as eligible group as they age from 60 - 69.

IV. Results

Looking at our results we see that there is insignificant evidence for the causal effect of OAS eligibility causing a decrease in labour force participation. In Table 2 we see our model differentiate between the effects of aging and the impact of OAS eligibility.

A variable we see is our treated_age variable. This shows us that individuals who are in the 65 to 69 group are 27.2 percentage points less likely to stay in the labour force, regardless of OAS. Using this variable we are able to see the overall retirement trend when not impacted by OAS.

Looking at the interaction between is_eligible and treated_age we see that this interaction was statistically insignificant($\beta_3 = 0.008, p > .1$), this tells us there is not enough evidence supporting that OAS is causing a drop in the labour force market. Overall strengthening the fact that the overall drop was predominantly caused by age rather than OAS.

After we added our demographic and categorical factors, we see a shift of our R^2 from 0.077 to 0.229. This is telling us our covariates like education and household income are a significant portion of the variation in employment status. Even with these factors accounting for 22.9% variance of the reason why people work. This shows us that regardless of these covariates, OAS eligibility potentially provides additional reason to exit the labour force.

TABLE 2—REGRESSION RESULTS: IMPACT OF OAS ON LABOUR FORCE PARTICIPATION

	Dependent variable:		
	in_labour_force		
	(1)	(2)	(3)
is_eligible	-0.045^{***}	-0.048^{***}	-0.067^{***}
	(0.005)	(0.005)	(0.004)
treated_age	-0.283^{***}	-0.284^{***}	-0.272^{***}
	(0.006)	(0.006)	(0.006)
is_eligible:treated_age	0.010	0.011	0.008
	(0.007)	(0.007)	(0.006)
Demographic controls	No	Yes	Yes
Categorical controls as factors	No	No	Yes
Observations	126,446	126,446	126,446
R^2	0.077	0.091	0.229
Adjusted R^2	0.077	0.091	0.227
Note:			*p<0.1; **p<0.05; ***p<0.01

V. Discussion + Extension

One threat to our claim is unobserved heterogeneity. There are many factors that the Census data does not provide information for. This include health, job satisfaction, job loss out of individuals control, or owning a paid off housing and other assets. This causes a threat of immigrants having to work even with OAS eligibility because they do not have the same assets as Canadians who have lived in Canada their whole lives and own homes. This could mean our result may skew the effect of OAS if the control group is overall likely to be worse off financially.

An assumption of our model is that we assume the impact of categorical controls are constant across all ages. An example is education, where we believe that education will keep us in the workforce longer, due to the factors of white collar work being much less labour intensive. However we do not account for other factors that are outweigh the effect of education; age, health, family or burnout. As we age there are factors like health or family. Education may just be a factor that accounts for 10% why an individual is in work force for a 68 year old, who cares more about family and is less career focused. While it may be 100% for a 60 year old as they might be much more career focused and healthy. This shows us that this change may not be as constant as we think.

To further extend our model we could use a fuzzy Regression Discontinuity Design to address our limitations. Our running variable would be age, instrument would be age >= 65 else 0 and variable being OAS benefits. This would allow us to see individuals who are the same in terms of health or education, but different just based on the amount gotten from the government. This would allow RDD to account for issues like our heterogeneity issues.

VI. Conclusion

We see there is insignificant evidence that there is a causal effect of OAS eligibility causing a decrease in labour force participation. To improve and extend this paper I would get more data and factors like wealth of individuals, or current macroeconomic status and do an analysis over multiple years rather than just one dataset for 1 year. Using this I would also look more into how it differs for each person and base my results off that. I believe this is a good extension for this paper as we would see a truer effect in the sense that we would be able to see the changes from the factors that make a difference. I also lack the use of CPP, to extend even further I could use CPP and see how much of the drop is due to CPP vs OAS or a combination of both.

REFERENCES

Blackshaw, Matthew, and Elizabeth Cahill. 2026. "Canada's Retirement Income System." *Library of Parliament.* Accessed April 10, 2026. https://lop.parl.ca/sites/PublicWebsite/default/en_CA/ResearchPublications/201940E.

Employment and Social Development Canada. 2026. "Old Age Security: Do you qualify." *Government of Canada.* Accessed April 8, 2026. https://www.canada.ca/en/services/benefits/publicpensions/old-age-security/eligibility.html.

Employment and Social Development Canada. 2026. "Old Age Security: Recovery Tax." *Government of Canada.* Accessed March 31, 2026. https://www.canada.ca/en/services/benefits/publicpensions/old-age-security/recovery-tax.html.

Immigration, Refugees and Citizenship Canada. 2026. "What Does It Mean to Have Legal Status in Canada?" Accessed February 25, 2026. https://ircc.canada.ca/english/helpcentre/answer.asp?qnum=515&top=15.

Stick, Max, Christoph Schimmele, Stephane Arabackyj, Jianwei Zhong, and Feng Hou. 2026. "Trends in the wealth gap between immigrant and Canadian-born families from 2016 to 2023." *Statistics Canada.* Accessed March 25, 2026. https://www150.statcan.gc.ca/n1/pub/36-28-0001/2026003/article/00002-eng.htm.